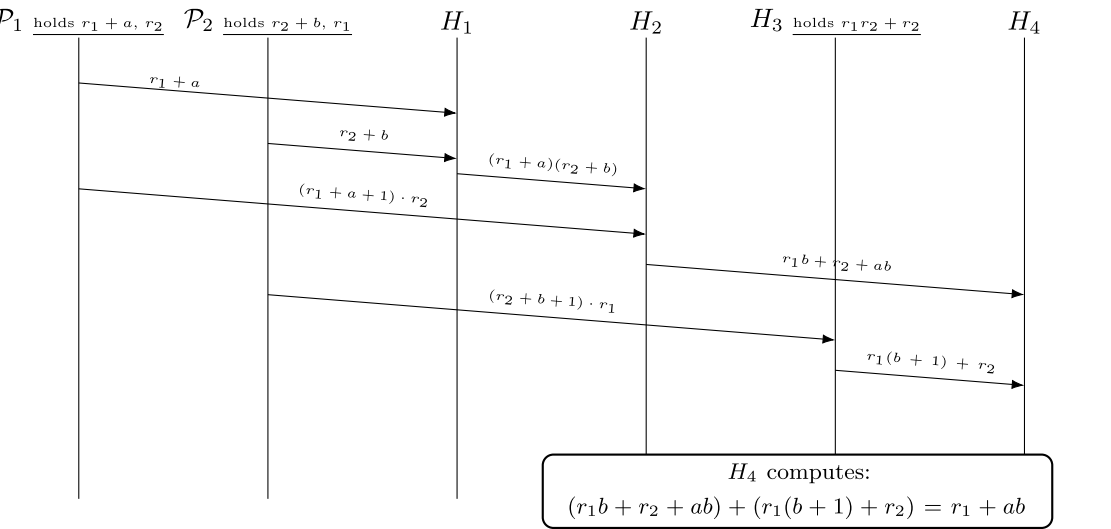
$\mathcal{P}_1$ holds $r_1 + a$, r_2
$\mathcal{P}_2$ holds $r_2 + b$, r_1
H_1
H_2
H_3 holds $r_1 r_2 + r_2$
H_4
$r_1 + a$
$r_2 + b$
$(r_1 + a)(r_2 + b)$
$(r_1 + a + 1) \cdot r_2$
$r_1 b + r_2 + ab$
$(r_2 + b + 1) \cdot r_1$
$r_1(b + 1) + r_2$
H_4 computes:
$(r_1 b + r_2 + ab) + (r_1(b + 1) + r_2) = r_1 + ab$